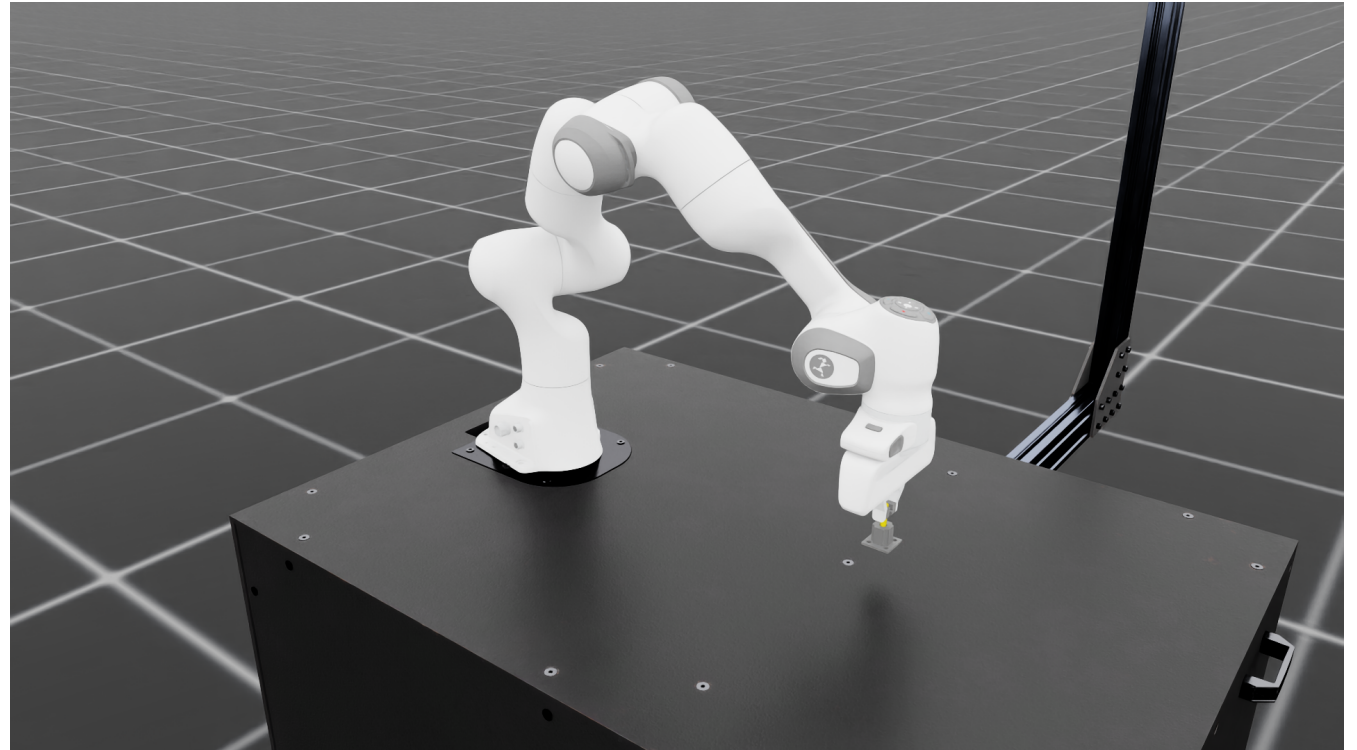


Safe insertion and recovery under load limits

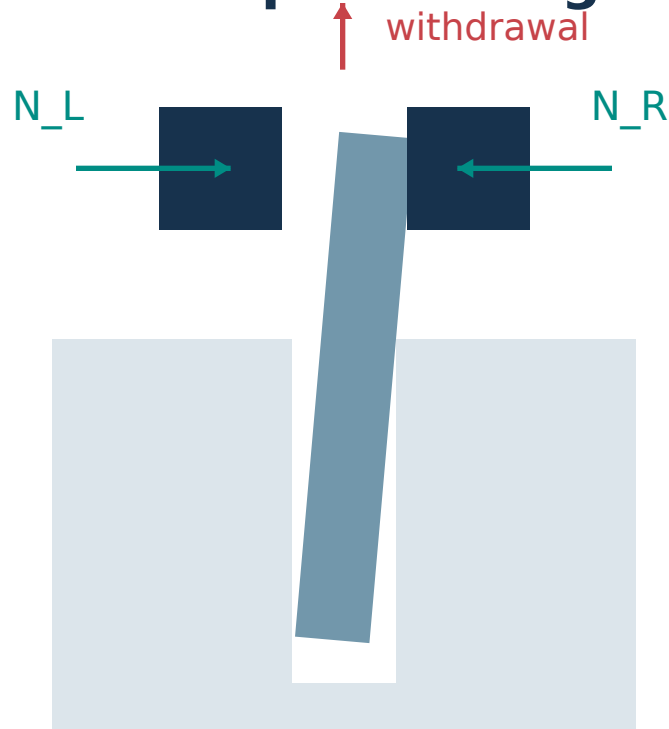
A simulation proof of concept
for task-aware recovery

Research progress and technical roadmap
September 2026



The task needs an affordable exit

Insert while preserving a feasible recovery



Peg and socket cutaway
Schematic, clearance enlarged

Limited grip friction

$$F_{t,max} \approx \mu_g (N_L + N_R)$$

Fragile parts also limit squeeze and bending.

Current simulation constraint

$$\|F_{wrist}\| \leq B_F \quad \|\tau_{wrist}\| \leq 1 \text{ N}\cdot\text{m}$$

3 / 4 / 5 N are exploratory task budgets.

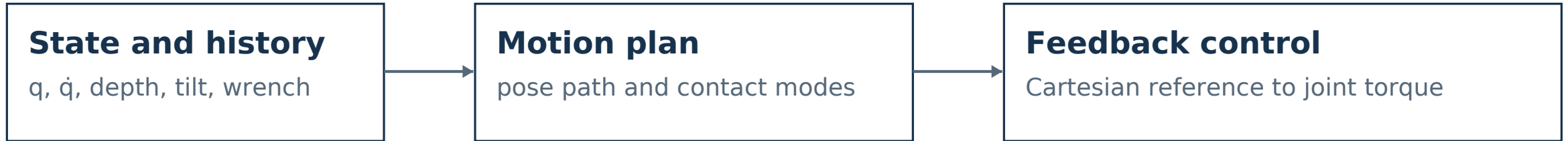
Grip capacity and fracture limits need calibration.

Research context

Contact and extraction	Geometry, friction and compliance shape mating forces.	Whitney 1982 [1] Goli et al. 2024 [2]
Interaction control	Virtual stiffness and damping shape contact response.	Hogan 1985 [3]
Planning through contact	Trajectory optimization can include changing contact modes.	Posa et al. 2014 [4]
Safety and transfer	Predictive filtering and calibrated deployment address model risk.	Wabersich 2021 [8] IndustReal 2023 [6]

Our focus: budget-conditioned recovery with matched insertion histories.

A constrained planning and control problem



Kinematics and interaction control

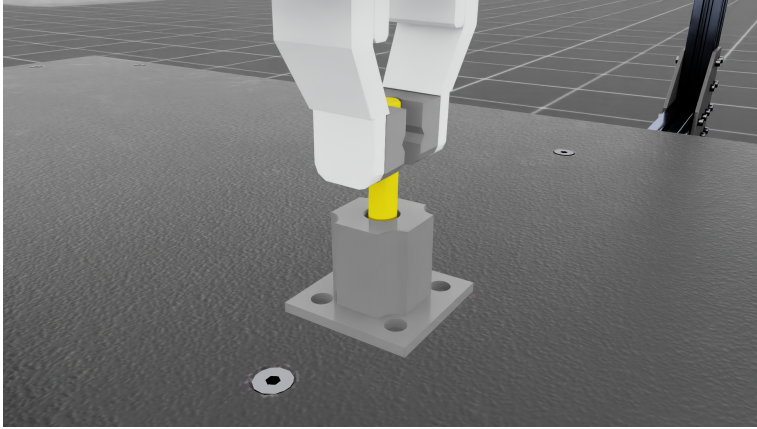
$$\tau = J(q)^T (K_p e - K_d \dot{x}) + \tau_{\text{null}}$$

Existing task-space PD law, simplified. e is the position/orientation error.

Planning objective Minimize recovery time and control effort
subject to load limits, joint limits and retained grasp.

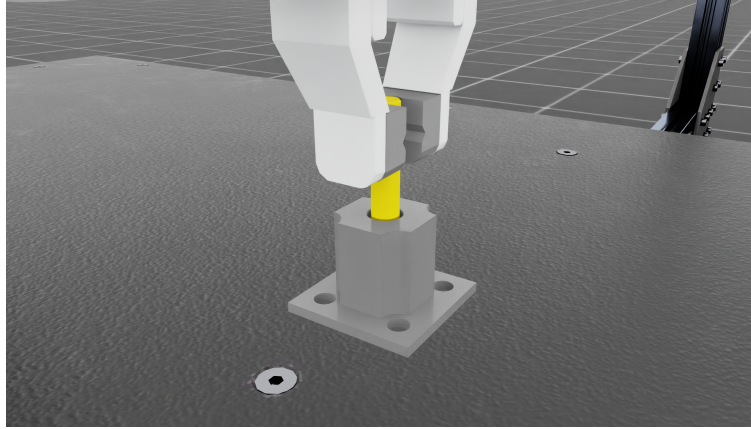
Policy-specific recoverability: can this path reach clearance within budget?

The simulated experiment



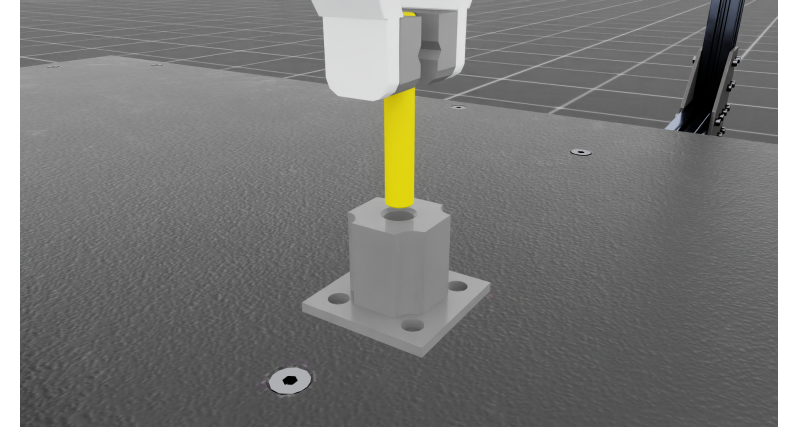
Reference state

18.228 mm depth, 1.439° tilt



Recentered and realigned

18.076 mm depth, 0.170° tilt



Clearance

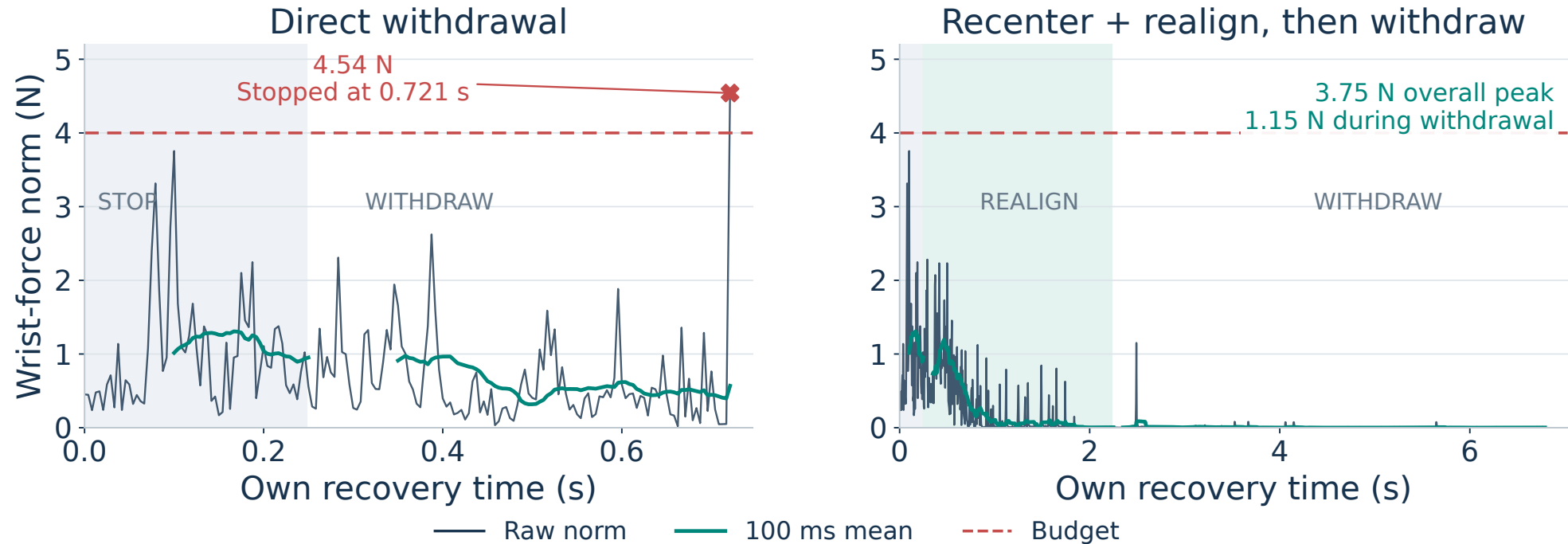
-3.009 mm depth, 0.063° tilt



Rigid bodies, 0.2 mm radial gap, 25 mm blind hole. Physics and task control: 240 Hz.

A recovery choice changes the outcome

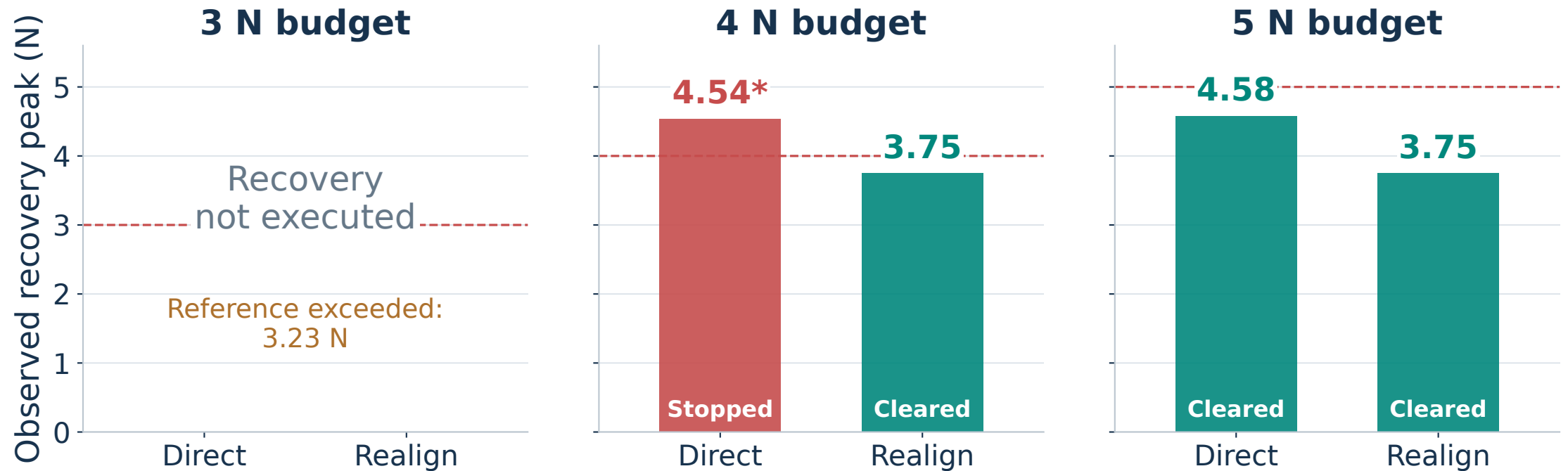
18 mm target depth · $\mu = 1$ · 1.5° command tilt · 4 N budget



Matched reference: 2,811 samples per policy · reference peak 3.56 N.

Direct path: first exceedance, incomplete. Realignment: complete recovery from the matched starting state.

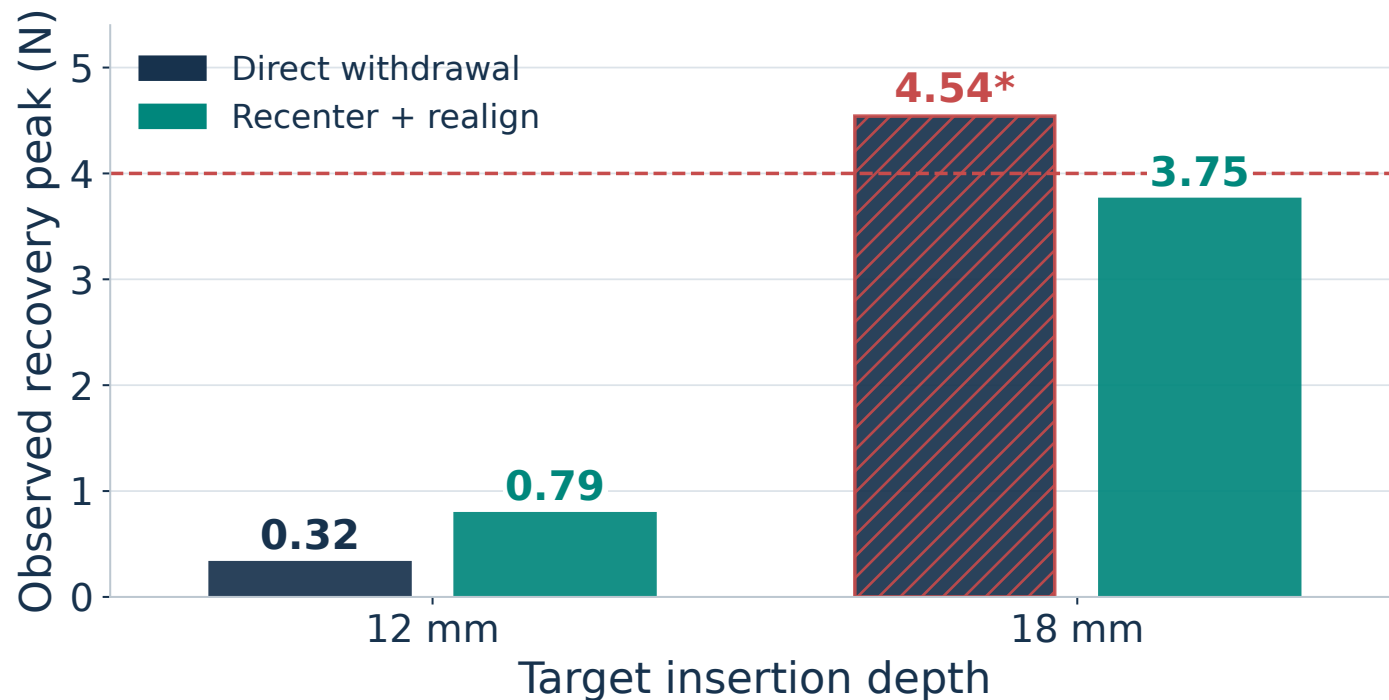
The budget changes feasibility



* First-exceedance stop; complete direct-withdrawal peak remains unobserved.

Recoverability depends on state, candidate policy and the task budget.

Embedding depth changes the recovery cost



* Censored at budget exceedance. Actual initial tilt: 1.52° / 1.44°.

18 mm direct path

1.105 mm

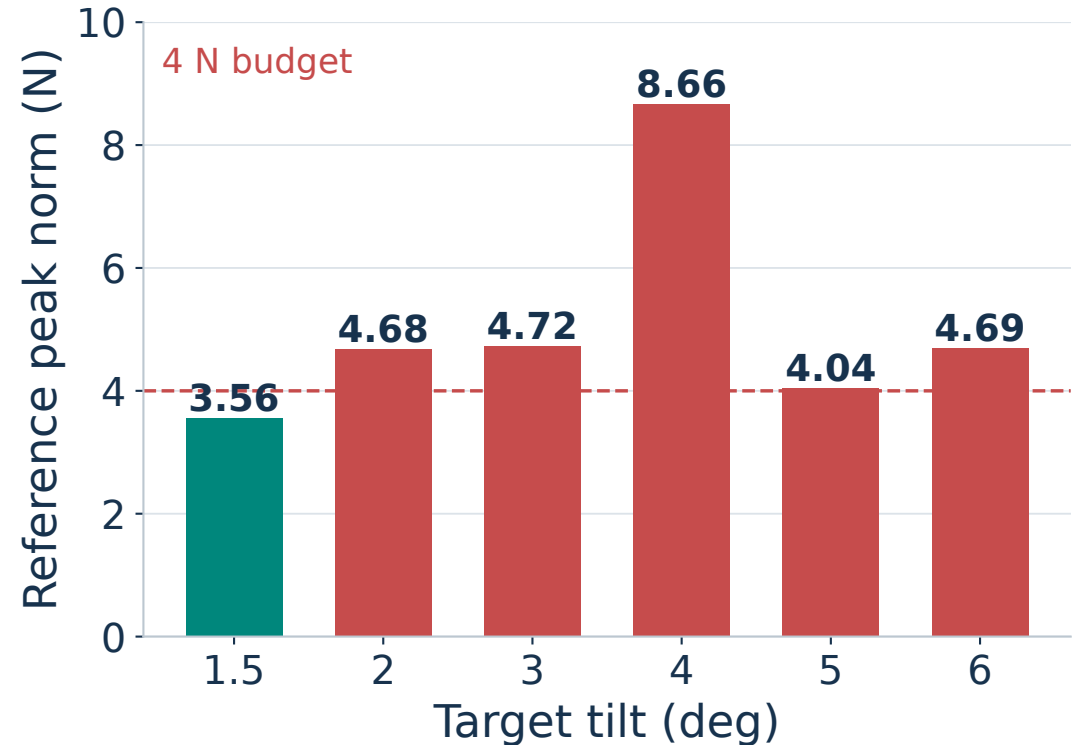
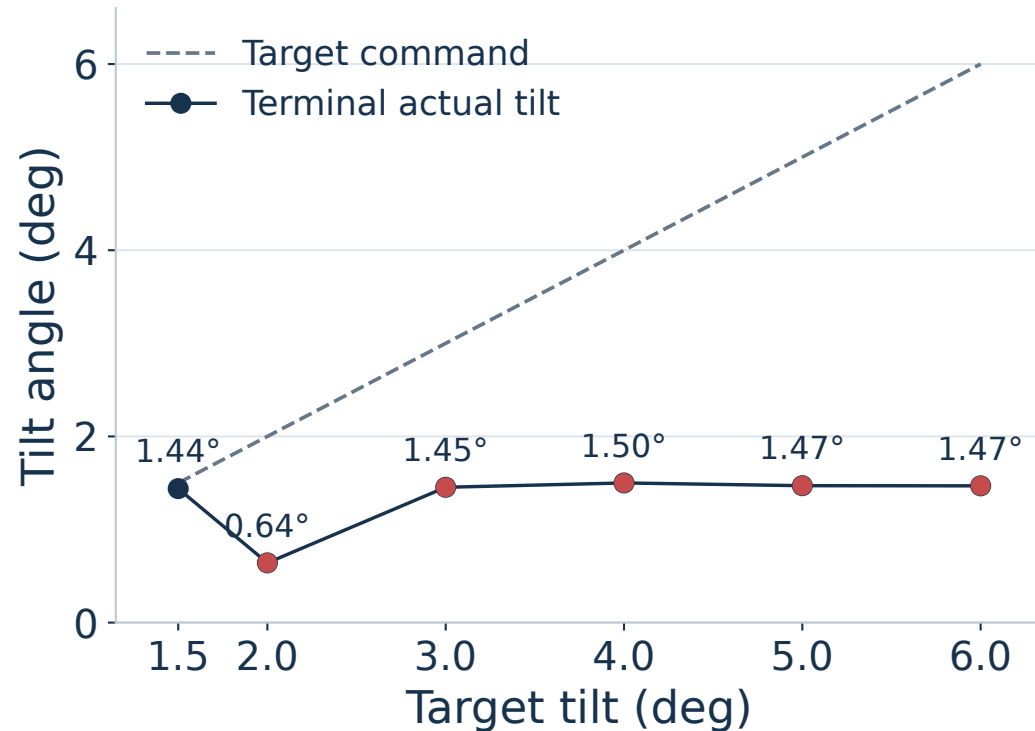
Commanded withdrawal
at the stopping frame

0.137 mm

Measured withdrawal
at the same frame

The adjustment motion has a cost, so realignment is not always preferable.

A 6° command does not produce a 6° state

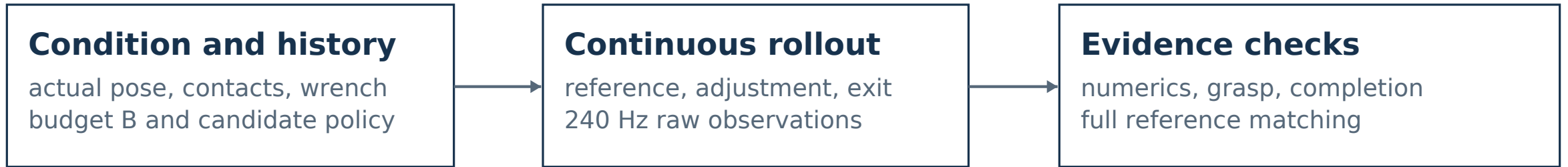


2°–6° commands: stopped during tilt, before withdrawal. Fixed 1 s ramp also changes angular speed.

6° target: 3.743° last command, 1.469° actual tilt, 4.694 N at stopping.

Fixed 1 s ramp also changes angular speed. These branches never entered recovery.

A dataset for recovery decisions



Safe clearance

Complete exit within budget

Load violation

Keep the first crossing frame

Grasp limit

Keep the separate grasp flag

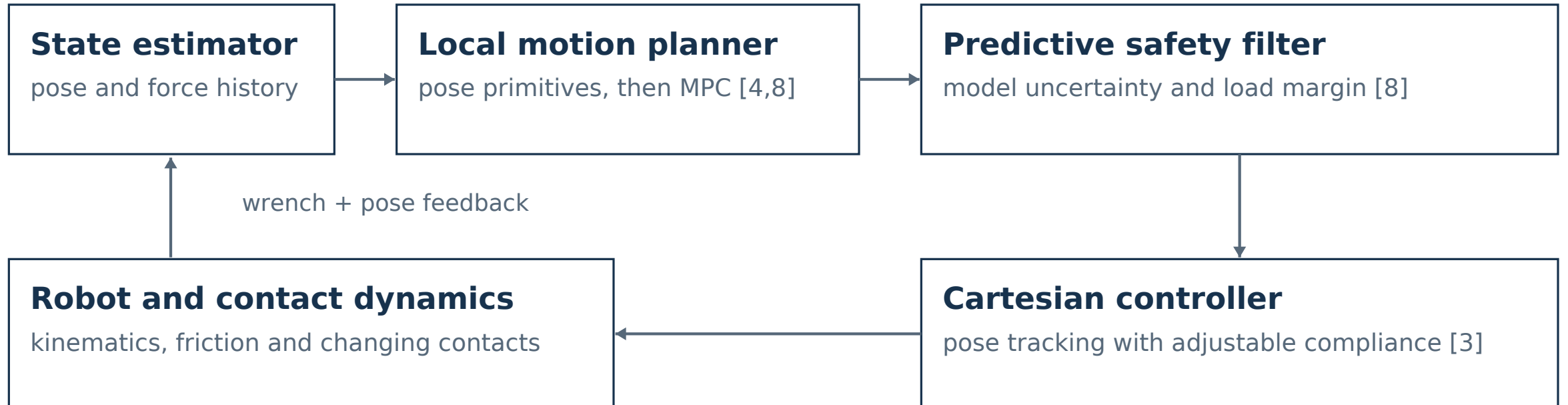
Unknown / untested

Missing or invalid evidence

Assign recovery labels only after a valid reference.
Keep unobserved trajectory segments missing.

Group shared physical paths together when splitting training and test data.

A proposed planning and control loop



MPC: model predictive control over depth, lateral offset, tilt and speed.

Evaluate predicted peak load and a feasible backup path before applying the next reference.

Control barrier functions (CBFs) [5] require validated dynamics and feasible safety constraints.

Feasibility is established at the concept level

Demonstrated

Matched state + 4 N budget:
 direct withdrawal exceeds, realignment clears.

01 Validate the simulator

Separate physics/control rates.
 Check solver, speed and parameter sensitivity.

02 Develop predictive recovery

Compare direct, realign and local MPC.
 Evaluate unseen path/parameter groups.

03 Calibrate and transfer

Measure grip and allowable part loads.
 Validate metal-peg / plastic-socket trials.

Concept verified in simulation. Robustness and hardware validation are next.

Complete evidence coverage

Depth	μ	Tilt	Budget	Direct withdrawal	Recenter + realign
18 mm	1	0°	4 N	Clear within budget	Clear within budget
18 mm	1	1°	4 N	Clear within budget	Clear within budget
18 mm	1	1.5°	4 N	Exceeded in withdrawal	Clear within budget
18 mm	1	1.5°	5 N	Clear within budget	Clear within budget
12 mm	1	1.5°	4 N	Clear within budget	Clear within budget
18 mm	0.5	1.5°	4 N	Exceeded in reference	Not tested
18 mm	1	2°	4 N	Exceeded in reference	Not tested
18 mm	1	1.5°	3 N	Exceeded in reference	Not tested

[D1] Budget study: 8 conditions, 13 executed branches, 3 untested.

[D2] Angle extension: 6 conditions, 7 executed branches, 5 untested.

Exploratory deterministic trials. Shared histories and repeated anchors are not independent samples.

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Characterizing the mechanics of rectangular peg-hole disassembly and the effect of the active compliance centre on the extraction force.

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[3] Hogan, N. (1985)

Impedance Control: An Approach to Manipulation: Part I - Theory.

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[4] Posa, M., Cantu, C., and Tedrake, R. (2014)

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[5] Ames, A. D., Xu, X., Grizzle, J. W., and Tabuada, P. (2017)

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[6] Tang, B., et al. (2023)

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[8] Wabersich, K. P., and Zeilinger, M. N. (2021)

A predictive safety filter for learning-based control of constrained nonlinear dynamical systems.

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